

Linear Regression from Scratch

Subject: Machine Learning Topic: Supervised Learning

Overview

Mathematical derivation, gradient descent implementation, and evaluation

Model and loss

- For one feature, linear regression predicts $\hat{y} = w \cdot x + b$; for multiple features, $\hat{y} = Xw + b$.
- Mean squared error is the mean of (prediction - target) squared and penalizes large errors strongly.
- Fit parameters by minimizing the loss on training data, then evaluate on held-out data.

Gradient descent

- For MSE with n examples, gradient for w is $(2/n)$ times X -transpose multiplied by prediction error; gradient for b is $(2/n)$ times the sum of errors.
- Update each parameter by subtracting `learning_rate` times its gradient.
- A learning rate that is too high can diverge; one that is too low converges slowly.

Practical workflow

- Standardize features when scales differ greatly to make gradient descent better conditioned.
- Add an intercept term or keep b as a separate parameter.
- Plot training loss across iterations; a steadily falling loss is a useful sanity check.

Evaluation and assumptions

- MAE is the mean absolute error; RMSE is the square root of MSE and remains in target units.
- R-squared compares model error with predicting the target mean; it is not proof of causality.
- Inspect residuals for nonlinearity, changing variance, outliers, and leakage between training and test data.